

JADEN ORLI

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RESEARCH FOCUS

Marine ecologist focused on the population dynamics of harvested species and the habitats that support them. Experience spans stage-structured modeling of California spiny lobster under Marine Protected Area and slot limit scenarios, fishery-independent surveys across four long-term California monitoring programs, and experimental coral biology. Interested in making monitoring data more useful for stock assessment and spatial management.

EDUCATION

University of California, Santa Barbara, Santa Barbara, CA

2025

Degree: BA in Biology (College of Creative Studies)

Awards: Regent Scholar, Coastal Fund Major Grant Recipient

GPA: 3.90

WORK EXPERIENCE

Stier Ocean Recoveries Lab, Research Intern & Staff Research Associate I, UCSB, CA 09/2024 – current

- Developing a stage-structured population model in R to evaluate how Marine Protected Areas (MPAs) and slot limits affect California spiny lobster fisheries
- Contributing to an island-wide *Porites* genotyping survey across Mo'orea reef sites, assigning colonies to mitochondrial lineages
- Independently designed and am running a 2x2 factorial experiment on genotyped *Porites* cores, testing whether secondary wounding and algae removal reactivate stalled tissue regeneration through GFP/RFP fluorescence imaging, amplicon metabarcoding of marginal algae, ImageJ quantification, and survival modeling in R
- Designed and built the saltwater cultivation system supporting ongoing research on coral fragment recovery under varying environmental parameters at UCSB

SONGS Mitigation Monitoring Program, Laboratory Assistant III, UCSB, CA 08/2025 – 01/2026

- Conducted intensive fishery-independent field surveys in Southern California salt marshes, sampling fish and invertebrate communities for the San Onofre Nuclear Generating Station (SONGS) Mitigation Monitoring Program
- Processed field samples in the lab, including species identification to lowest practical taxon and QA/QC data entry; operated and maintained field and lab equipment across challenging conditions including muddy salt marshes
- Collaborated with project scientists on sample tracking, data management, and reporting deliverables supporting long-term restoration monitoring

Valentine Lab, Research Intern, UCSB, CA 01/2023 – 08/2025

- Investigated the role of Marine Group II Archaea in biogenic hydrocarbon cycling and petroleum spills
- Applied bioinformatics and metagenomics pipelines using Bash (Terminal) and the Bridges-2 high-performance computing environment (PSC) to analyze microbial community dynamics
- Collected sediment and water samples on multi-day oceanographic cruise expeditions, including the March 2023 coring cruise aboard the research vessel Yellowfin that recovered the deep coastal sediment behind the group's DDT waste inventory

Casele Lab (PISCO), Scientific Diver/Research Intern, UCSB, CA 06/2024 – 10/2024

- Completed standardized underwater benthic and fish visual surveys as part of the MPA monitoring project with the Partnership for Interdisciplinary Studies of Coastal Oceans (PISCO) across the Northern Channel Islands
- Assisted with MARINe (Multi-Agency Rocky Intertidal Network) biodiversity surveys, conducting point contact, quadrat, and swath transects to characterize intertidal community composition across California sites
- Evaluated nearshore fish stocks through fishery-independent hook-and-line surveys with volunteer anglers via the California Collaborative Fisheries Research Project (CCFRP)

Reef Check California, Intern, CA

06/2022 – 04/2023

- Entered survey data from scientific divers at Reef Check monitoring sites across the coastal western United States
- Completed scientific diving transects in Southern California to measure abundance and prevalence of target invertebrate and kelp species
- Performed QA/QC review on surveys entered in the Reef Check California database

Novex Innovations, Lab Technician, Winston-Salem, NC

08/2020 – 08/2021

- Supported research and development of a drug for lymphatic malformation (historically manufactured in Japan)
- Passaged and processed mesenchymal stem cells from amnion and chorion tissue samples for research into regenerative therapies
- Trained new team members on sterile technique protocols (ISO 5) and proper equipment usage
- Participated in biweekly management meetings with department heads

LEADERSHIP EXPERIENCE

Kuris Parasite Ecology Lab, Project Lead, UCSB, CA

09/2021 – 03/2026

- Led a team of undergraduate students on an independent project investigating a novel copepod egg predator infesting commercially important rock crabs (*Cancer* spp.) in Santa Barbara
- Mentored peers to develop posters and presentations for various conferences including:
 - EEMB URS (2021/2022/2023): talks and posters at the UCSB Ecology, Evolution, and Marine Biology (EEMB) Undergraduate Research Symposium (URS)
 - ESA (2023): a talk at the 108th annual meeting of the Ecological Society of America (ESA)
- Authored and received a Coastal Fund Major Grant to fund student salaries and purchase research equipment (light boxes, drawing tube, Canon camera, and scope adapter)
- Provided multi-year field and laboratory support for graduate dissertation research spanning several California coastal ecosystems, surveying sites, laying transects, monitoring camera traps, and collecting and processing samples
- First-authored peer-reviewed publication in Ecology on the copepod egg predator (Orli et al. 2025; <https://doi.org/10.1002/ecy.70263>)

Environmental Media Production Course, Associate Producer, UCSB, CA

03/2025 – 04/2025

- Assisted with media management responsibilities associated with the course, including client communications and media organization
- Edited a short promotional film for a nonprofit organization, which collaborated with the Bren graduate school to host a climate leadership program for high school students, using Adobe Premiere Pro
- Filmed and directed high school students outdoors with a Canon EOS C70 camera
 - [Climate Leadership Program - Santa Cruz Island 2025](#)

VOLUNTEER EXPERIENCE

Santa Barbara Coastal Long Term Ecological Research, Scientific Diver, UCSB, CA

03/2023 – 05/2024

- Conducted underwater surveys in the kelp-forest monitoring surveys off Santa Barbara to track changes across multiple ecosystem components
- Deployed and retrieved packages of giant kelp at varying depths to characterize degradation rates under different physical conditions

Pacific Manta Research Group, Web Designer/Research Intern

12/2021 – 01/2024

- Designed the group's website with a focus on citizen-scientist photo-ID submission workflow
- Produced posters and presentations for divemasters and recreational divers covering manta etiquette and photo-identification technique
- Wrote a research proposal for the BBC Blue Planet III documentary on the significance of the Revillagigedo oceanic manta population

Floge Marine Microbial Ecology Lab, Research Intern, WF, NC

01/2020 – 03/2020

- Prepared petri dishes and growth mediums
- Pipetted samples and nutrients

- Worked with visitors to further enhance their experience through education
- Oversaw the visitor-animal engagement within the aquarium

SKILLS AND INTERESTS

- *Technical*: proficient in R, ArcGIS Pro, Adobe Premiere Pro, and WordPress; ecological modeling (stage-structured, spatial simulations, sensitivity analysis)
- *Projects*: spatially explicit analyses in GitHub
 - [Marine-Aquaculture](#): An analysis of potential locations within the Economic Exclusion Zones (EEZ) along the West Coast of the US that are potentially suitable locations for integrative multi-trophic aquaculture (IMTA)
 - [Houston-Storms](#): An analysis of blackout impacts during the Great Texas Freeze of 2021, evaluating socioeconomic disparities and spatial patterns of recovery across the Houston metropolitan area

RELEVANT COURSEWORK

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|--|--------------------------------------|---|
| • Applied Marine Ecology | • Microbial Diversity | • Biology of Fishes |
| • Theoretical Ecology | • Marine Ecology and Conservation | • Quantitative Methods in Biology |
| • Geospatial Analysis and Remote Sensing | • Ecological Modeling (Undergrad TA) | • Environmental Processes in Oceans and Lakes |
| • Environmental Modeling | • Environmental Media Production | • Marine Biogeochemistry |

DIVING EXPERIENCE

- *Total Dives*: ~ 450
- *Scientific Dives*: ~ 100
- *Certifications*: Rescue, Nitrox, AAUS (60 feet), and Reef Check
- *Conditions/Environments*: wrecks, low visibility, blue water, night, cold water, and strong currents
- *Technical Skills*: underwater photography (photo identification techniques), drift diving, and deep diving
- *Research Skills*: laying transects, conducting benthic and fish surveys, and invertebrate specimen collection

GRANTS

Coastal Fund, UCSB, CA (CF-202204-02367)

06/2022 – 03/2024

- *Major Grant Recipient*: received for “A novel copepod egg predator found infesting commercially important rock crabs in Santa Barbara, CA”
- *Author, Amount Awarded*: \$12,518.10

Coastal Fund, UCSB, CA (CF-202404-09827)

06/2024 – 06/2026

- *Major Grant Recipient*: received for “Microbial hydrocarbon cycling at Coal Oil Point”
- *Author, Amount Awarded*: \$8,100.00

AWARDS

Southern Academy of Parasitologists, Annual Meeting

05/2023

- *Best Student Presentation Award*: received for the poster titled “A novel copepod egg predator found infesting commercially important rock crabs in Santa Barbara, CA”

International Conference on Copepoda, Triannual Meeting

07/2022

- *Best Student Presentation Award*: received for the poster titled “A novel copepod egg predator found infesting commercially important rock crabs in Santa Barbara, CA”

Western Society of Naturalists, Annual Meeting

11/2021

- *Honorable Mention for Best Poster*: received for the poster titled “A novel copepod egg predator found infesting commercially important rock crabs in Santa Barbara, CA”

PUBLICATIONS (ORCID: 0009-0006-8419-6937)

Manuscripts:

- **Jaden E. Orli**, Sophia M. Lecuona, Gabrielle O. Plewe, Carson N. Gadler, Armand M. Kuris, Danny Tang, and Zoe L. Zilz. 2025. "Discovery of an Unidentified Species of Nicothoid Copepod Infesting Cancrid Crabs in Santa Barbara, California." *Ecology* 106(12):e70263. <https://doi.org/10.1002/ecy.70263>
- Zoe L. Zilz, Emily Hascall, Athena DiBartolo, Delén Flores, Sophie Cameron, **Jaden E. Orli**, and Armand M. Kuris. 2026. "A parasite-inclusive food web for the California rocky intertidal zone." *Scientific Data*. <https://doi.org/10.1038/s41597-026-07259-3>
- Mong Sin Christine, Wu, Jacob T. Schmidt, Hailie E. Kittner, **Earth 182B Group**, and David L. Valentine. 2025. "Geospatial and Temporal Inventory for Industrial DDT Waste Disposal to a Deep Coastal Ocean Environment." *Environmental Science and Technology* 59: 20578-20587. <https://doi.org/10.1021/acs.est.5c03851>

Acknowledged Contributions:

- Hailie E. Kittner. 2024. "Anthropogenic Structuring of Microbial Communities in Deep Coastal Sediment." *Master's thesis, University of California, Santa Barbara*. Sailed the March 2023 coring cruise aboard the research vessel Yellowfin that collected the sediment this thesis is based on.
- Zoe L. Zilz. 2024. "Connected by Tides: Missing Links in California's Coastal Ecotone." *Doctoral dissertation, University of California, Santa Barbara*. Provided field and laboratory support across several of the ecosystems in this dissertation over multiple years.